

i When to Use

To compare the means of two or more quantitative variables obtained from dependent samples (repeated measures or matched groups). The two or more scores might be the same variable measured at two or more different times or under different conditions, two or more comparable variables measured at the same time, or some combination.

H Research Hypothesis: Actors playing Clark Kent/Superman are older on average than actors playing Lois Lane across Superman media portrayals.

H Null Hypothesis: The mean ages of actors playing Clark Kent and Lois Lane in Superman portrayals are the same.

Research Design

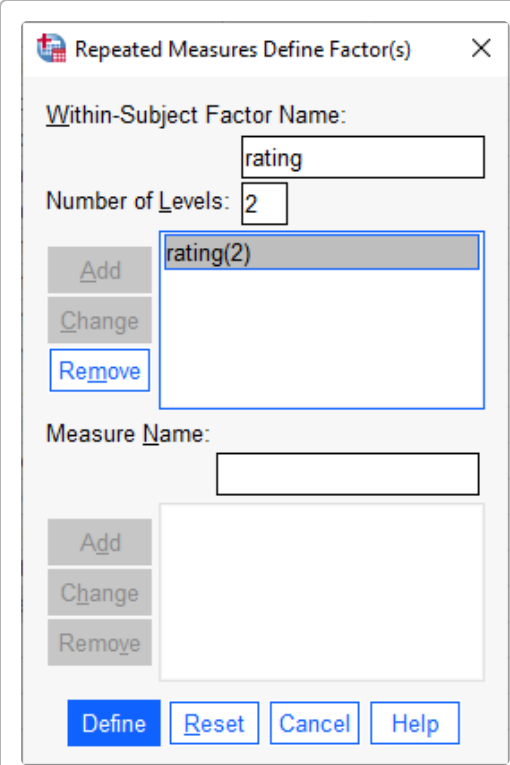
The IV is Character Role, with the conditions Clark Kent and Lois Lane. The DV is the actor's age at time of portrayal.

Clark Kent	Lois Lane

Variables in the Analysis: In a WG design the variables in the analysis are the variables holding the DV scores for each IV condition (clark_age & lois_age).

Analyze → General Linear Model → Repeated Measures

- In the **Repeated Measures Definition Window** enter your name for the IV in the “Within-subject Factor Name” box (CharRole)
- Enter the number of conditions of the IV in the “Number of levels” window (2)
- Click the “Add” button
- Click the “Define” button
- In the **Repeated Measures** window highlight the variables that are the DV score for each condition and click the arrow
- Click the “Options” button — in the **Repeated Measures: Options** window check the “Descriptives” box



Repeated Measures Define Factor(s)

Within-Subject Factor Name:

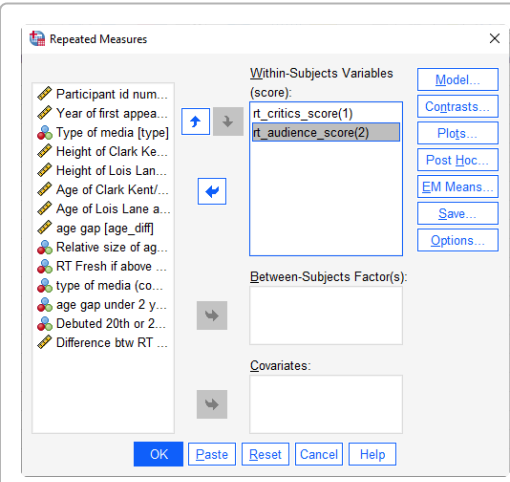
Number of Levels:

Add Change Remove

Measure Name:

Add Change Remove

Define Reset Cancel Help



Repeated Measures

Participant id num...
Year of first appea...
Type of media [type]
Height of Clark Ke...
Height of Lois Lan...
Age of Clark Kent/...
Age of Lois Lane a...
age gap [age_diff]
Relative size of ag...
RT Fresh if above ...
type of media (co...
age gap under 2 y...
Debuted 20th or 2...
Difference btw RT ...

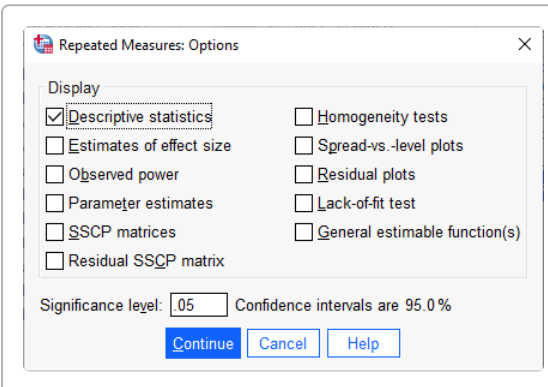
Within-Subjects Variables (score):

Model... Contrasts... Plots... Post Hoc... EM Means... Save... Options...

Between-Subjects Factor(s):

Covariates:

OK Paste Reset Cancel Help



Repeated Measures: Options

Display

☒ Descriptive statistics ☐ Homogeneity tests

☐ Estimates of effect size ☐ Spread-vs.-level plots

☐ Observed power ☐ Residual plots

☐ Parameter estimates ☐ Lack-of-fit test

☐ SSCP matrices ☐ General estimable function(s)

☐ Residual SSCP matrix

Significance level: Confidence intervals are 95.0 %

Continue Cancel Help

SPSS Syntax

— □ ×

GLM clark_age lois_age /WSFACTOR=CharRole 2 /PRINT=DESCRIPTIVE.

- ① List variables holding DV for each IV condition
- ② Give a name to the WG IV & tell # conditions
- ③ Get mean and std

SPSS Output

Descriptive Statistics

	Mean	Std. Deviation	N
clark_age	30.58	4.883	9
lois_age	30.17	6.408	9

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source	Type III Sum of Squares	df	Mean Square	F
Sig.				
CharRole Sphericity Assumed	0.728	1	0.728	0.027
.873 Error(CharRole) Sphericity Assumed	215.001	8	26.875	

SPSS provides different “versions” of the ANOVA output. We will focus on the “traditional” analysis, which SPSS labels as “Sphericity Assumed”.

- **df effect & F & p-value** to use
- **df(error) & Mean Square Error** term for this analysis

Assuming the null hypothesis is true, the p-value is the probability of obtaining an F statistic at least as large as the one observed. It is not the probability that the result is a Type I error.

SPSS may display **.000** when a very small p-value rounds to three decimal places. Report that value as **p < .001**, not p = .000.

Reporting the Results

Sample APA Write-Up

Figure/Table 1 shows the mean actor ages for Clark Kent and Lois Lane across 9 Superman portrayals. Clark Kent ($M = 30.58$, $SD = 4.88$) and Lois Lane ($M = 30.17$, $SD = 6.41$). There was no significant difference between the mean actor ages, $F(1,8) = 0.03$, $MSe = 26.88$, $p = .873$. Contrary to the research hypothesis, there was no significant difference between the mean ages of actors playing Clark Kent and Lois Lane.

It is important to report the univariate statistics for the dependent variable for both groups before presenting the ANOVA results. Often these are presented in a table or a figure.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- The statistical results
- Whether or not those results support the research hypothesis

Sometimes, the univariate statistics are presented in text, along with the ANOVA results.